Technical Report: Fast-Growing Firms Prediction

Data Analysis 3 - Assignment 2
Submitted by: Balint Decsi (ID: 2506626), Zariza Chowdhury (ID: 2500086)

February 2026

1 Introduction

This technical report details the methodology, modeling choices, and evaluation results for predicting fast-growing firms using the Bisnode panel data (2010-2015). The code and full notebooks are available in the associated GitHub repository: `https://github.com/zarizachow/Data-Analysis-3/tree/main`.

2 Data Preparation & Feature Engineering

2.1 Sample Design

We utilized the `bisnode-firms` dataset. The cleaning process involved:

- **Panel Construction:** We constructed a panel for 2010-2015.
- **Target Window:** The prediction target is defined over the 2012-2013 period.
- **Filtering:** We filtered for firms that were "alive" (positive sales) and had sales between 1,000 and 10 million EUR in the baseline year (2012).

2.2 Target Definition & Rationale

We define "fast growth" as a binary outcome where a firm achieves a sales growth rate in the top 10% of the distribution over a one-year horizon (2013 vs. 2012).

$$\text{Growth} = \ln(\text{Sales}_{2013}) - \ln(\text{Sales}_{2012})$$

Firms with growth $\geq 90^{th}$ percentile were labeled as `1` (fast-growing), and others as `0`.

Choice of Horizon (1-year vs. 2-year): We specifically chose a one-year growth horizon (2012-2013) over a two-year alternative (2012-2014) for several reasons rooted in corporate finance principles:

1. **Mean Reversion:** Corporate growth rates typically exhibit mean reversion—firms with exceptionally high growth in one period often see their growth slow down in subsequent periods as competition enters or market saturation occurs. A shorter one-year horizon captures the immediate, high-momentum phase we are interested in predicting, whereas a two-year horizon might dilute this signal with the natural decay of growth rates.
2. **Survivorship Bias:** Predicting growth over a longer two-year horizon requires firms to survive for an additional year. High-growth firms are often riskier and more volatile. By extending the horizon, we risk introducing survivorship bias, where the sample becomes skewed towards safer, more established firms, effectively filtering out the volatile high-flyers we aim to identify.

3. **Forecasting Relevance:** From a practical investment or credit scoring perspective, short-term forecasts are often more actionable. Financial planning and credit assessment cycles typically operate on an annual basis. A one-year prediction aligns better with these standard operational cycles than a medium-term two-year forecast.

2.3 Features

We engineered a rich set of features based on financial reports and firm characteristics:

- **Financials:** Log sales, assets, liabilities, profit/loss ratios.
- **Growth:** Lagged sales growth (`d1_sales_mil_log`).
- **Demographics:** Firm age, CEO age, region, industry category (`ind2_cat`).
- **Quality Flags:** Flags for data errors or extreme values in accounting fields.

Missing numeric values were imputed with the mean, and categorical values with the mode. One-hot encoding was applied to categorical variables.

The relationship between firm size and growth is notably non-linear, as seen below. This supports the use of tree-based models like Random Forest.

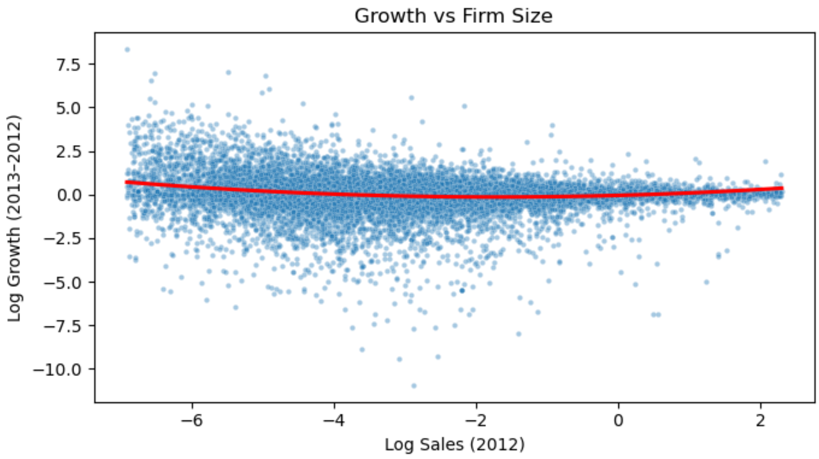

Figure 1: Growth vs. Firm Size (Log Sales)

3 Modeling Strategy

3.1 Models

We trained three models to predict the probability of fast growth:

1. **Logistic Regression (L2):** Standard parametric baseline.
2. **Logistic Regression (LASSO/L1):** For feature selection in high dimensions.
3. **Random Forest:** Non-parametric ensemble to capture non-linearities (e.g., Age vs. Growth).

3.2 Evaluation

We used 5-fold Stratified Cross-Validation.

- **Part I (Probability):** Metrics included ROC-AUC, Average Precision, and Brier Score.

- **Part II (Classification):** We defined a loss function with $Cost(FN) = 4 \times Cost(FP)$.

4 Results

4.1 Part I: Probability Prediction

The Random Forest model significantly outperformed the linear models.

Model	ROC-AUC	Avg Precision	Log Loss	Runtime (s)
Random Forest	**0.828**	**0.460**	**0.275**	~ 3.5
Logit (LASSO)	0.753	0.269	0.433	~ 742
Logit (L2)	0.753	0.269	0.433	~ 50

Table 1: Cross-validated performance metrics.

The Random Forest's superior performance (AUC 0.83 vs 0.75) highlights the importance of non-linear interactions in predicting firm growth.

4.2 Part II: Classification & Loss Minimization

We optimized the classification threshold t to minimize the expected loss:

$$\text{Expected Loss} = \frac{1}{N} \sum (FP \times 1 + FN \times 4)$$

- **Random Forest:** Minimized Loss = **0.255** at Threshold $\approx$ **0.15**.
- **Logit (L2):** Minimized Loss = 0.316 at Threshold ≈ 0.45.

The lower threshold for Random Forest reflects the high cost of false negatives; the model is calibrated to cast a wider net to catch the valuable fast-growers.

4.3 Part III: Confusion Matrix

Below is the confusion matrix for the Random Forest model (Fold 0) at the optimal threshold.

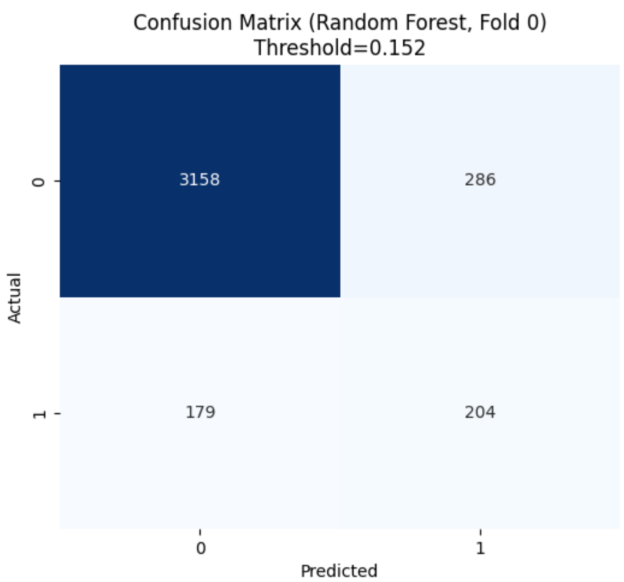

Figure 2: Confusion Matrix (Random Forest)

4.4 Part IV: Industry Analysis

We repeated the classification exercise for two subgroups:

1. **Manufacturing:** AUC = 0.817, Expected Loss = 0.239
2. **Services:** AUC = 0.838, Expected Loss = 0.257

The model maintains high performance across both sectors, validating its generalizability.

5 Code Snippet: Model Training

```python
# Example pipeline construction
preprocess = ColumnTransformer(
    transformers=[
        ("num", numeric_transformer, num_cols),
        ("cat", categorical_transformer, cat_cols),
    ]
)

rf_model = RandomForestClassifier(
    n_estimators=300,
    min_samples_leaf=10,
    random_state=42,
    n_jobs=-1
)

rf_pipe = Pipeline(steps=[("prep", preprocess), ("model", rf_model)])

# Cross-validation loop for loss minimization
for train_idx, test_idx in cv.split(X, y):
    rf_pipe.fit(X.iloc[train_idx], y.iloc[train_idx])
    probs = rf_pipe.predict_proba(X.iloc[test_idx])[:, 1]
    # ... threshold calculation ...
```

6 Conclusion

The Random Forest model is the clear choice for this prediction task. Its ability to model complex, non-linear relationships results in superior ranking capability (high AUC) and lower business risk (low Expected Loss).